Physical and impedence studies on PVdC-co-AN based gel polymer electrolyte

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***Abstract*— Lithium ion conducting membrane with PVdC-co-AN/$LiClO_4$ were prepared by carbonate (PC) solution casting technique. The preparation and physical and electrochemical properties of the gel polymer electrolytes have been briefly explicated in this paper. The conductivity increased upon increasing the polymer content and reduction of plasticizer content simultaneously. The maximum ionic conductivity has been found as 1.72×10^{-4} S/cm at 302 K for the (0.7) PVdC-co-AN-(0.2) $LiClO_4$-(0.1) PC electrolyte system. This may be due to the closeness of the free volumes in the polymer segmental sites. More plasticizer would results more dispersion which results in reduced conductivity.**

Keywords—Ionic conductivity, gel polymer electrolyte, plasticizer

I. INTRODUCTION

The science of polymer electrolytes is a vastly specified interdisciplinary field which includes the disciplines of electrochemistry, polymer science, organic chemistry, and inorganic chemistry. The field has involved ever-increasing interest, both in academia and industry, for the past two decades due to the potentially promising applications of such electrolytes, not only in all solid-state rechargeable lithium or lithium-ion batteries, but also in other electrochemical devices[1-3]. Except for PEO and its modifications, a number of polymer matrices which do not contain the $–CH_2\ CH_2\ O–$ repeating unit either in the backbone or in the side-chain have been plasticized to obtain electrolytes for high ambient-temperature conductivity.

Among polymer hosts such as PAN[4], poly vinyl pyrrolidinone PVP[5] , poly (vinylidene flouride-co-hexafluoropropylene) (PVdF-co-HFP)[6], and PAN-based polymer electrolytes have been extensively studied. The choice of the salt is actuated by the smaller size of the cations, larger size of anions and low dissociation energy. The lithium salts like LiX (X = $N(CF_3SO_2)_2$, BF_4,ClO_4, and CF_3SO_3) are resolute, solvable and dissociate easily and give high ionic conductivities. On the other hand, the addition of vinylidene chloride unit with poly (acrylonitrile) (PAN) gives high mechanical stability. The ionic conductivity of PAN-based electrolytes is of the order of 10^{-5}–10^{-6} S cm^{-1} with good thermal stability. PEO–oligomers with poly(vinylidene chloride-co-acrylonitrile)-based electrolytes have attained the conductivity in the order of 10^{-4}–10^{-5} S cm^{-1}[7]. In this work, to synthesize PVdC-co-AN with $LiClO_4$ based gel polymer electrolyte this could find applications in rechargeable lithium batteries.

II. EXPERIMENTAL

The polymer electrolytes were prepared by solution casting technique. Poly (vinylidene chloride) (PVDC) and poly (acrylonitrile) (PAN) were used as polymer matrix, lithium per chlorate ($LiClO_4$) as the doping salt, propylene carbonate (PC) as plasticizer was added. The polymer PVDC-CO-AN with salt $LiClO_4$ was first dissolved in tetrahydrofuran (THF) separately and then mixed together. The given amount of plasticizer was also added into the above polymer, salt mixtures. The polymer and salt was stirred continuously about 24 hours. The obtained homogeneous viscous was degassed to remove air bubbles and was degassed to remove air bubbles and was poured into the perti-dish. The solvent was allowed to evaporate slowly from the temperature at 500˚C for 12 hours. The polymer electrolyte films were arvested and stored in an evacuated desicater.

Materials characterization

The X'Pert PRO PANalytical X-ray diffractometer was used to do the structural analysis of the films at room temperature. The complex formation between the polymer complex and salt was ratified by SPECTRA RXI, PerkinElmer spectrophotometer in the range 400–4,000 cm-1. The ionic conductivity studies were carried out by placing the electrolyte film between the stainless steel blocking electrodes using a computer-controlled micro Autolab type III Potentiostat/
Galvanostat in the frequency range 100 Hz–300 KHz.

III. RESULTS AND DISCUSSION

Fig.1(a) shows that the X-ray diffraction pattern of pure polymer and salt. The x-raydiffraction of $LiClO_4$ reveals that the salt possessed hexagonal (Hexagonal structure and well matched with the literature). The polymer has also in almost amorphous nature. Fig.1(b) exhibits the x-ray diffraction patterns of the complexes comprising polymer lithium salt and different ratio of polymer plasticizer and fixed ratio of salt. These observations apparently reveal that the polymer undergoes significant structural reorganization upon the addition of plasticizer. The plasticizer might have induced significant disorder into the polymer, and this is

attributed to the interactions between polymer and the solvent, which resulted in polymer electrolytes with much lower crystallinity. Hence, the absence of peaks pertaining to lithium perchlorate salt in the complexes indicates the complete dissolution of the salt in the polymer and plasticizer. This result suggests that the incorporated lithium salt was dissolved in the amorphous region of polymer and the ionic conductivity of the polymer electrolyte mostly occurs in the amorphous phase of complex [8].

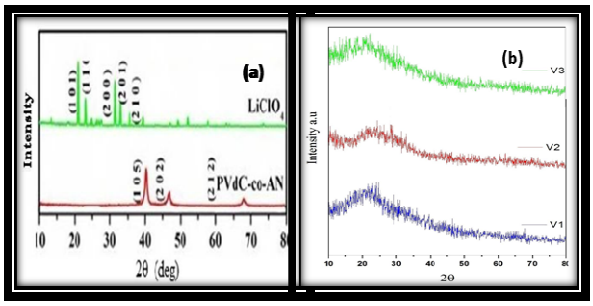

Fig.1.XRD patterns for (a) bare polymer and salt (b) as prepared samples

FT-IR spectroscopy is used to examine the interactions among the atom or ions of the polymer and salt in the electrolyte system. FT-IR spectra of PVdC-co-AN, $LiClO_4$ and its complexes with different ratio of plasticizer are shown in Fig.(2). The CH_2 asymmetric stretching, $C(Cl_2)$ symmetric stretching and the $C(Cl_2)$ bending of vinylidene chloride in pure PVdC-co-AN were observed in between ranges 1038-1073 cm^{-1}, 751–656cm^{-1} and at 568–446 cm^{-1} respectively[9]. However, in the electrolytes, these bands are found to be broader in complexes as salt concentration increases which is due to the structural re-organization of the polymer. The strong stretching vibrations of nitrile groups such as C≡N and C=N of PVdC-co-AN were appeared at 2247 and 1642 cm^{-1}, respectively in the complexes[10]. In acrylonitrile based electrolytes, the Li^+ ions associate with the nitrogen atoms of the nitrile group since the nitrile is a strong electron dopant[11].The presence of broad vibrational peak around 3600-3300cm^{-1} in the spectra corresponding to -OH group[10] is attributed to the water or moisture absorption which might have happened during the sample preparation for the spectral analysis. From the spectra, it is also noted that the vibrational bands of pure PVdC-co-AN are shifted from their original position, which confirm the complexation of the polymer with the salt.

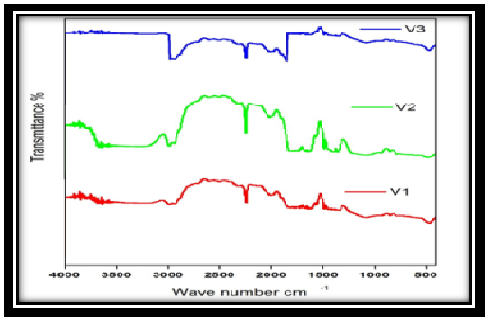

Fig.2.FTIR spectra of the as obtained samples

The surface morphology of the gel polymer electrolytes are given in Fig.3 (a-c) with the various magnifications. It is understood that the more dilution has been observed in the sample V1 than the rest. The plasticizer content reduction causes the little bit rough surface but, the optimized sample V3's conductivity is higher among all samples studied. It may be due to the dispersion of P(VdC-co-AN) has high elasticity which could be added with the high amount of plasticizer would results more diluted nature i.e., the polymer segments are more dispersed which will not provide more paths for lithium ions. Therefore, decreasing the PC content to 0.1 provides an optimized dispersion in the matrix and it makes the matrix more conduction. This could be revealed from the SEM images shown in Fig.3

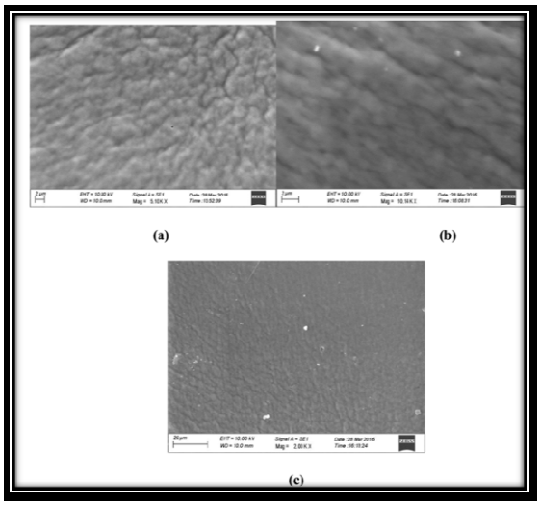

Fig.3.SEM images of the polymer complexes

The conduction mechanism is studied with the help of ac impedance spectroscopy by observing the contribution of polymeric chain, mobility, and carrier generation process [12]. Fig.4 (a-c) depicts the room temperature ionic conductivity of the prepared gel polymer electrolytes. This feature reveals that the conductivity in these films is established mainly by the ions and which is calculated using the formula σ=l/ (Rb•A), where A is the area of the film in contact with the electrodes, l is the thickness of the film, and the value of bulk resistance (Rb) is taken from the intercept of the impedance plots. The film V3 complexed with PC plasticizer is found to have maximum ionic conductivity value of 1.724x10-4Scm-1at 303 K. In general, the influence of plasticizer in the ionic conductivity of the polymer electrolyte films is determined by its own characteristics like dielectric constants, viscosity, polymer–plasticizer interaction, and ion–plasticizer interaction.

In this view, the plasticizer PC possesses a high dielectric constant (64).In the present case, conductivity increase with the increase of polymer content and decreased plasticizer content. This suggests that the plasticizer disperse the matrix and under reduced content of plasticizer, the dispersion

has been optimized as 10 wt. % in this case. The plasticizers would dissolve enough charge carriers and provide more mobile medium for the ions so as to enhance the conducting behaviour of the samples [13]. The ionic conductivity values of the sample observed over the temperature range 303–353 K have been plotted in Fig. 4(d).

It could be visualized that the conductivity values increase with the increase of temperature regardless the nature of the sample as given in Table.1. This enhancement in the conductivity values with increase of temperature could be elucidated with the free volume theory [14]. Free volume is produced when the polymer expands with temperature and the polymer segments of rmolecules salvated with ions can move spasmodically into the free volume. The free volume around the polymer chain decides the overall mobility of the ions and the polymer segment, which represents the resultant ionic conductivity of the films. Hence, the ionic and segmental mobility gear up the ion transport by overcoming the retarding effects of the ion clouds. The plot of the temperature dependent ionic conductivity indicates that the ion conduction in the gel polymer electrolytes.

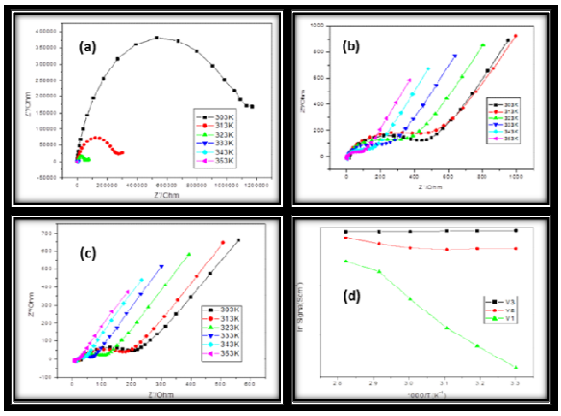

Fig.4(a-c).Impedance spectra of the as prepared samplesd(d)Ionic conductivity of the as prepared samples

The temperature dependent ionic conductivity measurements were carried out in the temperature range 303-353K and the plot fig 3.6 gives the variation of ionic conductivity with respect to temperature. Fig 4(d) suggests that the temperature dependent ionic conductivity plots seem to obey Arrhenius relation as given below and have been filted to obtain the parameters. $\sigma = \sigma 0 \exp(-Ea/KT)$ Where σ is the conductivity, Ea the activation energy associated with conduction, K the Boltzman constant and σ0 the pre-exponential term. From the Fig .4, the activation energy values (Ea) has been calculated from the slopes and are listed in theTable .1

Sample code	RT	313K	323K	333K	343K	353K	Activation energy
V1	3.847E-6	1.438E-6	5.177E-6	2.437E-6	1.303E-6	2.344E-6	1.3199
V2	5.099E-5	5.102E-5	4.842E-5	5.325E-5	6.884E-5	9.796E-5	0.7496
V3	1.724E-4	7.644E-4	2.199E-4	1.461E-4	1.205E-4	1.141E-4	0.6170

Table.1 Ionic conductivity and activation energy of the polymer complexesin the temperature range 303-353K

IV. CONCLUSION

The PVdC-co-AN-LiClO4 has been used as a host polymer and the propylene carbonate (PC) has been varied by fixing the $LiClO_4$ salt content for preparing the gel polymer electrolyte using solution casting technique. The complex formation in the polymer electrolytes has been as certained by studying the functional group vibration by FT-IR. Also, it could be realized through XRD analysis. The conductivity increased upon increasing the polymer content and reduction of plasticizer content simultaneously. The maximum ionic conductivity has been found as 1.72×10-4 S/cm at 302 K for the (0.7) PVdC-co-AN-(0.2) $LiClO_4$-(0.1) PC electrolyte system. This may be due to the closeness of the free volumes in the polymer segmental sites. More plasticizer would results more dispersion which results in reduced conductivity. SEM images also support this fact. Further study has been carried out by our group member to study its performance using composite particles incorporation in to the maximum conducting sample.

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